Registers

- fastest storage area
- inside CPU
- quickly accessable by CPU

X = Extended to 16 bits
E = Extended to 32 bits.

Registers

- General Purpose Register
  - Accumulator (A) (I/O operation) (ah, ax, aex)
  - Base (B) (holds address of data) (b, bx, ebx)
  - Counter (C) (counts, used in loops) (c, cx, ecx)
  - Data (D) (holds data for output) (d, dx, edx)
- Segment registers
  - Code segment (CS)
  - Data (DS) segment
  - Stack (SS) segment
  - Extra (ES) segment
- Index Registers
  - Source (SI) Index (points the source operand)
  - (DI) Destination Index (points the destination operand)
- Special Purpose Register
  - Instruction pointer (IP)/PC
    - holds the next instruction
  - Stack pointer (SP). points current top of sta

FLAG REGISTERS

(i) Carry Flag (CF) - Status
contain (0/1) carry from arithmetic operations. from LMB

(ii) Parity Flag (PF) - Status
indicates even or odd bit parity in the recieved data

Even parity
- if 1's is even = 0
- if 1's is odd = 1

Odd parity
- if 1's is even = 1
- if 1's is odd = 0

(iii) Auxiliary Flag (AF) - status
it's value would be one if there is carry after the third bit

(iv) Zero Flag (ZF) - status
Its value will be zero if the answer of the arithmetic operation is not zero. And its value will be 1 if the answer is 0.

(v) Sign Flag (SF) - Status.
if the value of the arithmetic operation is positive then SF = 0 and if its negative then its SF = 1.

(vi) Trap Flag (TF) - Control
TF = 1; activates a single step output
TF = 0; normal execution (default)

(vii) Interrupt Flag (IF) - Control.
If the IF = 1; the interrupt will be processed
If the IF = 0; the interrupt instruction will be ignored.

1) Hello

(viii) Direction Flag (DF) - Control

DF = 0 : text will be displayed from left to right

DF = 1 : text will be displayed from right to left.

(ix) Overflow Flag (OF) - Status

OF will be zero if the bits are not wasted or exceeded the 16 bit limit.

OF will be 1 if the bits are overflowed or exceeded 16 bits.

Hello World Program

```
.MODEL SMALL
.STACK 100H
.DATA
 msg DB 'Hello world$'
.CODE
 MAIN PROC
 MOV AX,@DATA
 MOV DS, AX.
 MOV DX,OFFSET msg
 MOV AH, 09h
 INT 21H
 MOV AH, 4Ch
 INT 21h
 MAIN 21h
MAIN ENDP
END MAIN.
```